CITY UNIVERSITY OF HONG KONG

EE5410 - Signal Processing
MATLAB Exercise 1
Telephone Touch-Tone Signal Encoding and Decoding

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2a) What is the time duration of `A`?

Ans: 0.3s

2b) How many elements are there in `A`?

Ans: 1201

2c) Modify toneA.m by changing "`F_A=440`" to "`F_A=800`". **Can you hear any difference? Describe the difference if any.**

Ans:
Yes, there are different.
The sound of F_A=440 is flatter than the sound of F_A=800.
The sound of F_A=800 is shriller (sharpen) than the sound of F_A=440.

3) Note that `tone` is a user-defined MATLAB function. Try the following commands: `help tone` and `y=tone(200,0.5)`. **What are the uses of these two commands?**

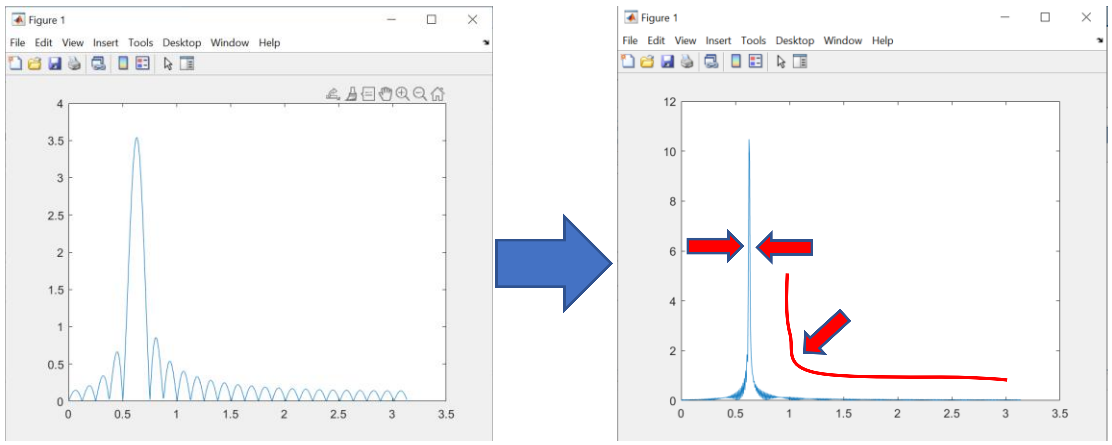
Ans:

1. `help tone` – display help information for the tone function. Because `tone` is a user-defined MATLAB function. So the comment (under `tone()` and start with % in the code) will be the help information

2. `y=tone(200,0.5)` – call `tone` function with frequency "200Hz" and the time duration of y "0.5s".

5a) What do you expect about the shape of the magnitude when $L \to \infty$?

Ans:



L=50 L=500

According to the magnitudes Plots by using L=50 and L=500, when L increase to 500, the width of main lobe decreases, area of side lobes become smaller, and the shape of side lobes become flat and smoother.

The shape of the magnitude should like a perpendicular line (⊥) when $L \to \infty$.

5b) Compute the energies of h[n] for L = 50 and L = 500

Ans:

For L = 50,
Energies E = 0.50

For L = 500,
Energies E = 0.50

where $\omega = 0.2\pi$

5c) Which filter will give a better DTMF decoding performance, h[n] with L = 50 or L = 500? Explain your answer.

Ans:
h[n] with L = 500 will give a better DTMF decoding performance because longer filter length can reduce the main lobe width and the transition width. Also it can decrease side lobes area and ripples.

Therefore, the DTMF decoding result become more precisely and perform well when h[n] with L = 500.

6. Try your `dtmfdetect` function with various keys, different L (L =50 and L =500) and noise powers (σ^2=0, σ^2=10 and σ^2=100). For each setting, perform 20 trials and record the number of correct detections in the following table.

Ans:

Key	L = 50 $\sigma^2 = 0$	L = 500 $\sigma^2 = 0$	L = 50 $\sigma^2 = 10$	L = 500 $\sigma^2 = 10$	L = 50 $\sigma^2 = 100$	L = 500 $\sigma^2 = 100$
1	**20**	**20**	**20**	**20**	**3**	**9**
2	**20**	**20**	**20**	**20**	**1**	**13**
3	**20**	**20**	**20**	**20**	**4**	**11**
4	**20**	**20**	**20**	**20**	**4**	**11**
5	**20**	**20**	**20**	**20**	**4**	**9**
6	**20**	**20**	**20**	**20**	**3**	**11**
7	**20**	**20**	**20**	**20**	**2**	**14**
8	**20**	**20**	**20**	**20**	**3**	**12**
9	**20**	**20**	**20**	**20**	**4**	**11**
0	**20**	**20**	**20**	**20**	**5**	**14**
*	**20**	**20**	**20**	**20**	**4**	**9**
#	**20**	**20**	**20**	**20**	**4**	**9**

-End-